

Bole Ma

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Research interests: Computer vision, geometric perception, and computational imaging.
Seeking Summer 2027 research and engineering internships.

Education

Purdue University Ph.D. in Electrical and Computer Engineering Advisor: Prof. Stanley H. Chan, Intelligent Imaging Lab	Sep 2024–Present; Expected May 2029
Shanghai Jiao Tong University M.S. in Electronic Information and Electrical Engineering	Sep 2021–Jun 2024
Shanghai Jiao Tong University B.S. in Electronic Information and Electrical Engineering	Sep 2017–Jun 2021

Technical Skills

Programming	Python, MATLAB, C++ (basic)
Deep Learning	PyTorch, neural-network training
Research Computing	Linux, Slurm, AI-assisted research workflows
Vision & Imaging	Geometric modeling, monocular depth estimation, camera calibration, optical system design and alignment
Experimental Methods	Dataset curation, experimental design, wavefront reconstruction

Research Experience

Geometry-Guided Monocular Metric Depth Estimation <i>Computer Vision · Purdue University</i>	Jan 2026–Present
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Developing an interpretable monocular metric depth framework for robotic and UAV perception, integrating explicit camera geometry with learned scene representations.

- Designed and implemented an interpretable depth-estimation framework integrating explicit camera geometry with learned scene representations.
- Built and ran the training pipeline, enabling metric depth prediction from monocular RGB images.
- Conducted controlled intervention experiments to assess how intermediate predictions influence depth outputs and guide ongoing model refinement.
- Curated task-specific training subsets from approximately 4.72 million TartanAir v2 frames, balancing scene coverage and camera-geometry diversity under a single-A100 compute budget.

DARPA CIDAR Challenge <i>Student Project Lead · Purdue University</i>	Jan 2025–Jun 2026 DARPA · Purdue News
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Passive image-based ranging for high-accuracy, low-latency distance estimation.

- Led data organization, analysis, and feasibility assessment of infrared-absorption, stereo-vision, and monocular depth-estimation approaches, guiding the team’s method selection.
- Collected approximately 1,000 UAV images, calibrated and organized the dataset, and trained a compact U-Net prototype; identified domain shift as a limitation of training solely on locally collected data.
- Developed a camera-geometry-based approach for passive image-based distance estimation, following feasibility assessment and neural-network prototyping.
- Helped Purdue place among the five semifinal awardees; the team received \$250,000 across white-paper and semifinal awards.

Pupil Design for Computational Wavefront Estimation

Jan 2026–Apr 2026

Computational Imaging · Purdue University

Investigating asymmetric pupil designs to resolve ambiguity in wavefront phase recovery from point spread functions (PSFs).

- Designed, built, and refined an optical testbed, including component selection, optical alignment, and troubleshooting.
- Captured 3,000+ experimental PSF images, of which 2,500 were selected for subsequent neural-network fine-tuning and wavefront reconstruction.

Selected Prior Research**Complex Wavefront Retrieval with a Higher Space–Bandwidth Product**

Sep 2022–Jun 2023

Computational Imaging · Shanghai Jiao Tong University

Master’s Research

Recovering complex wavefronts—including amplitude and phase—from diffraction intensity images while preserving field of view and spatial resolution.

- Built and trained an imaging-process-matched neural network that processes full-size diffraction patterns to reconstruct complex wavefronts with a larger field of view and higher spatial resolution.
- Designed the optical experimental setup, collected data, and validated reconstruction through simulations and physical experiments; wrote the first-author paper published in *Optics Letters* (2023).

Selected Publications

1. Weijian Zhang, Prateek Chennuri, Hashan K. Weerasooriya, **Bole Ma**, Stanley H. Chan. **Markov-Renewal Single-Photon LiDAR Simulator**. *European Conference on Computer Vision (ECCV)*, 2026.
2. Yu Yuan, Xijun Wang, Tharindu Wickremasinghe, Zeeshan Nadir, **Bole Ma**, Stanley H. Chan. **NewtonGen: Physics-Consistent and Controllable Text-to-Video Generation via Neural Newtonian Dynamics**. *International Conference on Learning Representations (ICLR)*, 2026.
3. Ali Almualllem, Nicholas Chimitt, **Bole Ma**, Qi Guo, Stanley H. Chan. **Pupil Design for Computational Wavefront Estimation**. *arXiv preprint*, 2026.
4. Xijun Wang, Xingguang Zhang, Yu Yuan, **Bole Ma**, Prateek Chennuri, Stanley H. Chan. **Generative Personalized Blind Face Restoration Enhanced by Physical Identity**. *IEEE International Conference on Image Processing (ICIP)*, 2025.
5. **Bole Ma**, Chuxuan Huang, Sibing Hou, Haiyu Zhang, Manhua Liu, Hao Yan. **Imaging process matched neural network for complex wavefront retrieval with a higher space–bandwidth product**. *Optics Letters*, 48(20), 5399–5402, 2023. **First author**.

Honors and Awards

- **DARPA CIDAR Challenge** — one of five semifinal awardees; Purdue team received \$250,000 across white-paper and semifinal awards, 2026.
- **Outstanding Graduate of Shanghai Municipality** — Top 5%.
- **Instrument 78 Motivation Fund**, Shanghai Jiao Tong University — Rank 1/52.
- **Endress+Hauser Scholarship**, Shanghai Jiao Tong University — Top 5%.
- **B-Grade Outstanding Scholarship**, Shanghai Jiao Tong University — Top 10%.